

Mensuration (10 th std)

Choose the Appropriate Answer

1. The curved surface area of a right circular cone of height 15 cm and base diameter 16 cm is _____

(a) $60\pi \text{ cm}^2$ (b) $68\pi \text{ cm}^2$ (c) $120\pi \text{ cm}^2$ (d) $136\pi \text{ cm}^2$

Ans: (d) $136\pi \text{ cm}^2$

$$l = \sqrt{h^2 + r^2} = \sqrt{15^2 + 8^2} = \sqrt{225 + 64} = \sqrt{289} = 17 \text{ cm}$$

$$\pi r l = \frac{22}{7} \times 8 \times 17 = 136\pi \text{ cm}^2$$

2. If two solid hemispheres of same base radius r units are joined together along with their bases, then the curved surface area of this new solid is

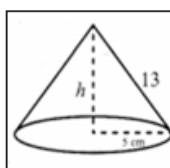
(a) $4\pi r^2$ sq. units (b) $6\pi r^2$ sq. units (c) $3\pi r^2$ sq. units (d) $8\pi r^2$ sq. units

Ans: (a) $4\pi r^2$ sq. units

3. The height of a right circular cone whose radius is 5 cm and slant height is 13 cm will be _____

(a) 12 cm (b) 10 cm (c) 13 cm (d) 5 cm

Ans: (a) 12 cm

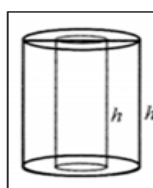


$$h = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm.}$$

4. If the radius of the base of a right circular cylinder is halved keeping the same height, then the ratio of the volume of the cylinder thus obtained to the volume of original cylinder is

(a) 1: 2 (b) 1: 4 (c) 1: 6 (d) 1: 8

Ans: (b) 1: 4



$$V_2 = \pi \left(\frac{r}{2}\right)^2 h = \frac{\pi r^2 h}{4}, \quad V_1 = \pi r^2 h = \pi r^2 h$$

$$\frac{V_2}{V_1} = \frac{\frac{\pi r^2 h}{4}}{\pi r^2 h} = \frac{\cancel{\pi} r^{\cancel{2}} h}{4} \times \frac{1}{\cancel{\pi} r^{\cancel{2}} h} = \frac{1}{4} = 1:4$$

5. The total surface area of a cylinder whose radius is of its height is

(a) $\frac{9\pi h^2}{8}$ sq.units (b) $24\pi h^2$ sq.units (c) $\frac{8\pi h^2}{9}$ sq.units (d) $\frac{56\pi h^2}{9}$ sq.units

Ans: (c) $\frac{8\pi h^2}{9}$ sq.units

$$\begin{aligned}
 \text{TSA of a cylinder} &= 2\pi r(h+r), r = \frac{1}{3}h \\
 &= 2\pi \frac{1}{3}h \left(h + \frac{1}{3}h \right) = \frac{2}{3}\pi h^2 + \frac{2}{3}\pi \frac{h^2}{3} = \frac{2}{3}\pi h^2 + \frac{2}{3}\pi \frac{h^2}{3} \\
 &= \frac{6\pi h^2 + 2\pi h^2}{9} = \frac{8\pi h^2}{9}
 \end{aligned}$$

6. In a hollow cylinder, the sum of the external and internal radii is 14 cm and the width is 4 cm. If its height is 20 cm, the volume of the material in it is _____

- (a) $560\pi \text{ cm}^3$ (b) $1120\pi \text{ cm}^3$ (c) $56\pi \text{ cm}^3$ (d) $360\pi \text{ cm}^3$

Ans: (b) $1120\pi \text{ cm}^3$

$$\begin{aligned}
 \text{Volume} &= \pi(R^2 - r^2)h = \pi(R+r)(R-r)h \\
 &= \pi(14)4 \times 20 = 1120\pi \text{ cm}^3
 \end{aligned}$$



7. If the radius of the base of a cone is tripled and the height is doubled then the volume is

- (a) Made 6 times (b) Made 18 times
(c) Made 12 times (d) Unchanged

Ans: (b) Made 18 times

$$\begin{aligned}
 v &= \frac{1}{3}\pi(3r)^2 \times (2h) = \frac{1}{3} \times \pi 9r^2 \times 2h \\
 &= 6\pi r^2 h = 18 \times \frac{1}{3}\pi r^2 h = \text{made 18 times.}
 \end{aligned}$$



8. The total surface area of a hemisphere is how many times the square of its radius

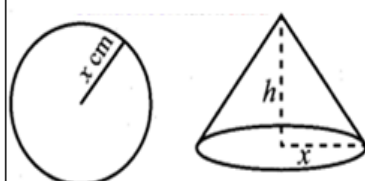
- (a) π (b) 4π (c) 3π (d) 2π

Ans: (c) 3π

9. A solid sphere of radius x cm is melted and cast into a shape of a solid cone of same radius. The height of the cone is

- (a) 3x cm (b) x cm (c) 4x cm (d) 2x cm

Ans: (c) 4x cm



$$\begin{aligned}
 \frac{1}{3}\pi r^2 h &= \frac{4}{3}\pi r^3 \\
 \frac{1}{3}\pi x^2 h &= \frac{4}{3}\pi x^3 \Rightarrow h = 4x \text{ cm}
 \end{aligned}$$

10. A frustum of a right circular cone is of height 16 cm with radii of its ends as 8 cm and 20 cm. Then, the volume of the frustum is _____
 (a) $3328\pi \text{ cm}^3$ (b) $3228\pi \text{ cm}^3$ (c) $3240\pi \text{ cm}^3$ (d) $3340\pi \text{ cm}^3$

Ans: (a) $3328\pi \text{ cm}^3$



$$\begin{aligned}
 V &= \frac{\pi h}{3} [R^2 + r^2 + Rr] \\
 &= \pi \times \frac{16}{3} [20^2 + 8^2 + 20 \times 8] \\
 &= \pi \times \frac{16}{3} [400 + 64 + 160] \\
 &= \pi \times \frac{16}{3} \times 624 = 3328\pi \text{ cm}^3
 \end{aligned}$$

11. A shuttlecock used for playing badminton has the shape of the combination of
 (a) a cylinder and a sphere (b) a hemisphere and a cone
 (c) a sphere and a cone (d) Frustum of a cone and a hemisphere

Ans: (d) Frustum of a cone and a hemisphere

12. A spherical ball of radius r_1 units is melted to make 8 new identical balls each of radius r_2 units. Then $r_1 : r_2$ is
 (a) 2: 1 (b) 1: 2 (c) 4: 1 (d) 1: 4

Ans: (a) 2: 1

Volume of the first sphere: Volume of second sphere = 8: 1

$$\frac{4}{3} \pi r_1^3 : \frac{4}{3} \pi r_2^3 = 8:1 \Rightarrow r_1^3 : r_2^3 = 2^3 : 1^3 \Rightarrow r_1 : r_2 = 2:1$$

13. The volume (in cm^3) of the greatest sphere that can be cut off from a cylindrical log of wood of base radius 1 cm and height 5 cm is ____
 (a) $\frac{4}{3} \pi$ (b) $\frac{10}{3} \pi$ (c) 5π (d) $\frac{20}{3} \pi$

Ans : (a) $\frac{4}{3} \pi$

14. The height and radius of the cone of which the frustum is a part are h_1 units and r_1 units respectively. Height of the frustum is h_2 units and the radius of the smaller base is r_2 units. If $h_2 : h_1 = 1: 2$ then $r_2 : r_1$ is
 (a) 1: 3 (b) 1: 2 (c) 2: 1 (d) 3: 1

Ans: (b) 1: 2

Ratio of their volumes $= \frac{1}{3} \pi h_1 (r_1^2 + r_2^2 + r_1 r_2) : \frac{1}{3} \pi h_2 (r_1^2 + r_2^2 + r_1 r_2)$

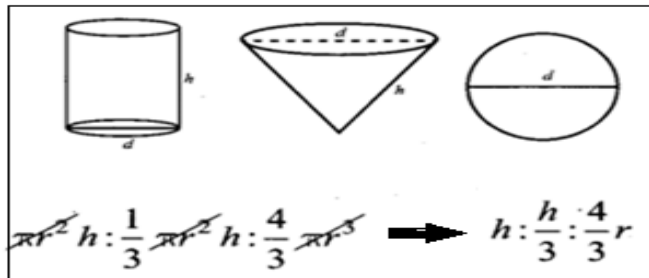
$$= 2 (r_1^2 + r_2^2 + r_1 r_2) : 1 (r_1^2 + r_2^2 + r_1 r_2)$$

$r_1 : r_2 = 2: 1$ But $r_2 : r_1 = 1: 2$

15. The ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height is _____

- (a) 1: 2: 3 (b) 2: 1: 3 (c) 1: 3: 2 (d) 3: 1: 2

Ans: (d) 3: 1: 2



16. The shape of an ice-cream cone is a combination of:

- (a) Sphere + Cylinder (b) Sphere + Cone
(c) Hemisphere + Cylinder (d) Hemisphere + Cone

Ans: (d) Hemisphere + Cone

17. If a cone is cut parallel to the base of it by a plane in two parts, then the shape of the top of the cone will be a:

- (a) Sphere (b) Cube (c) Cone itself (d) Cylinder

Ans: (c) Cone itself

18. If we cut a cone in two parts by a plane parallel to the base, then the bottom part left over is the:

- (a) Cone (b) Frustum of cone (c) Sphere (d) Cylinder

Ans: (b) Frustum of cone

19. If r is the radius of the sphere, then the surface area of the sphere is given by;

- (a) $4 \pi r^2$ (b) $2 \pi r^2$ (c) πr^2 (d) $\frac{4}{3} \pi r^2$

Ans: (a) $4 \pi r^2$

20. If we change the shape of an object from a sphere to a cylinder, then the volume of cylinder will

- (a) Increase (b) Decrease (c) Remains unchanged (d) Doubles

Ans: (c) Remains unchanged

21. Fifteen solid spheres are made by melting a solid metallic cone of base diameter 2cm and height 15cm. The radius of each sphere is:

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{3}\sqrt{2}$ (d) $\frac{1}{3}\sqrt{4}$

Ans: (d) $\frac{1}{3}\sqrt{4}$

$$\begin{aligned} \text{Volume of 15 spheres} &= \text{Volume of a cone} \Rightarrow 15 \times \frac{4}{3} \pi r^3 = \frac{1}{3} \pi r^2 h \\ 5 \times 4 \pi r^3 &= \frac{1}{3} \pi 1^2 (15) \Rightarrow 20r^3 = 5 \Rightarrow r^3 = \frac{5}{20} = \frac{1}{4} \Rightarrow r = \frac{1}{3}\sqrt{4} \end{aligned}$$

22. The radius of the top and bottom of a bucket of slant height 35 cm are 25 cm and 8 cm. The curved surface of the bucket is:

- (a) 4000 sq.cm (b) 3500 sq.cm (c) 3630 sq.cm (d) 3750 sq.cm

Ans: (c) 3630 sq.cm

Curved surface of bucket = $\pi(R_1 + R_2) \times \text{slant height (l)}$ Curved Surface = $(22/7) \times (25 + 8) \times 35$ CSA = $22 \times 33 \times 5 = 3630 \text{ sq.cm.}$
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23.If a cylinder is covered by two hemispheres shaped lid of equal shape, then the total curved surface area of the new object will be

- (a) $4\pi rh + 2\pi r^2$ (b) $4\pi rh - 2\pi r^2$ (c) $2\pi rh + 4\pi r^2$ (d) $2\pi rh + 4\pi r$

Ans: (c) $2\pi rh + 4\pi r^2$

24.A tank is made of the shape of a cylinder with a hemispherical depression at one end. The height of the cylinder is 1.45 m and radius is 30 cm. The total surface area of the tank is:

- (a) 30 m (b) 3.3 m (c) 30.3 m (d) 3300 m

Ans: (b) 3.3 m

Total surface area of tank = CSA of cylinder + CSA of hemisphere $= 2\pi rh + 2\pi r^2 = 2\pi r (h + r)$ $= 2 \times 22/7 \times 30(145 + 30) \text{ cm}^2 = 33000 \text{ cm}^2 = 3.3 \text{ m}^2$
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25.If we join two hemispheres of same radius along their bases, then we get a;

- (a) Cone (b) Cylinder (c) Sphere (d) Cuboid

Ans: (c) Sphere

26.A cylindrical pencil sharpened at one edge is the combination of

- (a) a cone and a cylinder (b) frustum of a cone and a cylinder
(c) a hemisphere and a cylinder (d) two cylinders

Ans: (a) a cone and a cylinder

27.A shuttle cock used for playing badminton has the shape of the combination of

- (a) a cylinder and a sphere (b) a cylinder and a hemisphere
(c) a sphere and a cone (d) Frustum of a cone and a hemisphere

Ans: (d) Frustum of a cone and a hemisphere

28.A solid piece of iron in the form of a cuboid of dimensions 49 cm × 33 cm × 24 cm, is moulded to form a solid sphere. The radius of the sphere is

- (a) 21 cm (b) 23 cm (c) 25 cm (d) 19 cm

Ans: (a) 21 cm

Volume of cuboid = volume of sphere moulded $49 \times 33 \times 24 = \frac{4}{3} \pi r^3 \Rightarrow \pi r^3 = 29106 \Rightarrow r^3 = (21)^3 \Rightarrow r = 21 \text{ cm}$
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29.A right circular cylinder of radius r cm and height h cm ($h > 2r$) just encloses a sphere of diameter

- (a) r cm (b) 2r cm (c) h cm (d) 2h cm

Ans: (b) 2r cm

30.Two identical solid cubes of side a are joined end to end. Then the total surface area of the resulting cuboid is

- (a) $12a^2$ (b) $10a^2$ (c) $8a^2$ (d) $11a^2$

Ans: (b) $10a^2$

31. A solid cylinder of radius r and height h is placed over another cylinder of same height and radius. The total surface area of the shape so formed is

- (a) $4\pi rh + 4\pi r^2$ (b) $2\pi rh + 4\pi r^2$ (c) $2\pi rh + 2\pi r^2$ (d) $4\pi rh + 2\pi r^2$

Ans: (d) $4\pi rh + 2\pi r^2$

32. The number of shots each having diameter 3 cm can be made from a cuboidal lead solid of dimensions 9 cm $\times$ 11 cm $\times$ 12 cm is approximately equal to

- (a) 84 (b) 90 (c) 92 (d) 80

Ans: (a) 84

Volume of cuboidal lead solid = 9 cm $\times$ 11 cm $\times$ 12 cm = 1188 cm³
 Radius of lead shot = 3 / 2 cm = 1.5 cm
 Volume of each shot = $(4/3) \pi r^3 = (4/3) \times (22/7) \times 1.5 \times 1.5 \times 1.5 = 14.143$ cm³
 Number of lead shots can be made = 1188 / 14.143 = 84 (approx.)

33. Two identical solid hemispheres of equal base radius r cm are stuck together along their bases. The total surface area of the combination is

- (a) $6\pi r^2$ (b) $5\pi r^2$ (c) $4\pi r^2$ (d) $3\pi r^2$

Ans: (c) $4\pi r^2$

34. A funnel is in the shape of a right circular cone with a base radius of 3 cm and a height of 4 cm. Find the slant height of the funnel.

- (a) 4 cm (b) 5 cm (c) 7 cm (d) 7.5 cm

Ans: (b) 5 cm

35. What is the total surface area of an iron sphere having a radius of 11 cm?

- (a) 1529.76 cm² (b) 1514.76 cm² (c) 1519.76 cm² (d) 1419.76 cm²

Ans: (c) 1519.76 cm²

The total surface area of an iron sphere = $4\pi r^2 = 4 \times 3.14 \times 11^2 = 1519.76$ cm²

36. What is the volume of a hemisphere if the radius of the hemisphere is 3 m?

- (a) 135.4 m³ (b) 56.52 m³ (c) 120.23 m³ (d) 105.5 m³

Ans: (b) 56.52 m³

Radius of the hemisphere = $(2/3) \times 3.14 \times 3^3 = 56.52$ m³

37. The lateral surface area of a right circular cone is 2020 cm² and its radius is 10 cm. What is its slant height?

- (a) 28.3 cm (b) 25.6 cm (c) 68.23 cm (d) 64.33 cm

Ans: (d) 64.33 cm

$\pi r l = 2020 \Rightarrow 3.14 \times 10 \times l = 2020 \Rightarrow l = 64.33$ cm

38. Find the lateral surface area of a rectangular room with a height of 22 m, length 27 m and a breadth of 23 m.

- (a) 4793 m² (b) 2263 m² (c) 2200 m² (d) 4783 m²

Ans: (c) 2200 m²

The lateral surface area of a cuboid = $2 h (l + b) = 2 \times 22 (27 + 23) = 2200$ m²

39. Find the total surface area of a square-shaped box having a length of 2 m for its side.

- (a) 47 m^2 (b) 22 m^2 (c) 24 m^2 (d) 48 m^2

Ans: (c) 24 m^2

$\text{The total surface area of a cube} = 6a^2 = 6 \times 2^2 = 24 \text{ m}^2$
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40. A solid is in the form of a cone mounted on a hemisphere. The radius and height of the cone are 3 m and 4 m. Find the surface area of the given solid.

- (a) 114.4 m^2 (b) 103.62 m^2 (c) 70 m^2 (d) 72.5 m^2

Ans: (b) 103.62 m^2

41. What is the length of the resulting solid if two identical cubes of side 8 cm are joined end to end?

- (a) 26 cm (b) 16 cm (c) 21 cm (d) 14 cm

Ans: (b) 16 cm

$\text{Length of resulting cuboid} = 2 \times \text{side of the cube} = 2 \times 8 \text{ cm} = 16 \text{ cm}$
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42. What is the formula to find the height of the tank consisting of a circular cylinder with a hemisphere attached on either end?

- (a) Radius of the cylinder + 2 (height of the hemisphere)
 (b) Height of the cylinder + 2 (height of the hemisphere)
 (c) Radius of the cylinder + 2 (radius of the hemisphere)
 (d) Height of the cylinder + 2 (radius of the hemisphere)

Ans: (d) Height of the cylinder + 2 (radius of the hemisphere)

43. What is the C.S.A of resulting solid if two identical cubes are joined end to end together with the length of the sides of the cube is 4 m?

- (a) 160 cm^2 (b) 96 cm^2 (c) 168 cm^2 (d) 604 cm^2

Ans: (b) 96 cm^2

Cuboid is the resulting solid when two identical cubes are joined end to end together. Length of the cuboid (l) = $4+4 = 8 \text{ cm}$ Height (h) = 4 cm Breadth (b) = 4 cm The curved surface area of cuboid = $2 h (l + b) = (2 \times 4) (8 + 4) = 96 \text{ cm}^2$
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44. What is the formula required to use for T.S.A of an article which is made by digging out a hemisphere from each end of a solid cylinder?

- (a) C.S.A of the cylinder – 2 (C.S.A of the hemisphere)
 (b) C.S.A of the cylinder + C.S.A of the hemisphere
 (c) C.S.A of the cylinder + 2 (C.S.A of the hemisphere)
 (d) C.S.A of the cylinder – C.S.A of the hemisphere

Ans: (c) C.S.A of the cylinder + 2 (C.S.A of the hemisphere)

45. A surahi is the combination of _____

- (a) a sphere and a cylinder (b) a hemisphere and a cylinder
 (c) Two hemispheres (d) a cylinder and a cone

Ans: (a) a sphere and a cylinder

46. If two solid hemispheres of the same base radius 'r' are joined together along their bases, then curved surface area of this new solid is
 (a) $4\pi r^2$ (b) $6\pi r^2$ (c) $3\pi r^2$ (d) $8\pi r^2$

Ans: (a) $4\pi r^2$

47. The radius (in cm) of the largest right circular cone that can be cut out from a cube of edge 4.2 cm is
 (a) 4.2 (b) 2.1 (c) 8.1 (d) 1.05

Ans: (b) 2.1

Diameter of base of largest possible cone = 4.2 cm $\text{Radius} = \frac{4.2}{2} = 2.1 \text{ cm}$
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48. How many bags of grain can be stored in a cuboid granary 12 m × 6 m × 5 m. If each bag occupies a space of 0.48 m³?
 (a) 750 (b) 75 (c) 1500 (d) 375

Ans: (a) 750

$\begin{aligned} \text{No. of bags} &= \frac{\text{volume of granary}}{\text{volume of one bag}} \\ &= \frac{12 \times 6 \times 5}{0.48} = 750. \end{aligned}$
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49. In a swimming pool measuring 90 m × 40 m, 150 men take a dip. If the average displacement of water by a man is 8 m³, then rise in water level is
 (a) 27.33 cm (b) 30 cm (c) 31.33 cm (d) 33.33 cm

Ans: (d) 33.33 cm

$\begin{aligned} \text{Volume of water displaced} &= 150 \times 8 = 1200 \text{ m}^3 \\ \Rightarrow 90 \times 40 \times h &= 1200 \\ \Rightarrow h &= \frac{1200}{90 \times 40} \text{ m} = 33.33 \text{ cm} \end{aligned}$

50. The shape of a Gilli, in the Gilli Danda game is a combination of
 (a) Two cylinders (b) a cone and a cylinder
 (c) Two cones and a cylinder (d) Two cylinders and a cone

Ans: (c) Two cones and a cylinder

51. During conversion of a solid from one shape to another, the volume of the new shape will _____
 (a) Increase (b) Decrease (c) Remain unaltered (d) Be doubled

Ans: (c) Remain unaltered

52. The shape of a glass tumbler is usually in the form of _____
 (a) A cone (b) Frustum of a cone (c) a cylinder (d) a sphere

Ans: (b) Frustum of a cone

53. In a right circular cone, the cross-section made by a plane parallel to the base is a

- (a) Circle (b) Frustum of a cone (c) Sphere (d) Hemisphere

Ans: (a) Circle

54. Which of the following is the correct formula for the volume of a sphere?

- (a) $(4/9) * \pi * r^3$ (b) $(2/9) * \pi * r^3$ (c) $(4/3) * \pi * r^3$ (d) $(2/3) * \pi * r^3$

Ans: (c) $(4/3) * \pi * r^3$

55. What will be the radius of a spherical vessel having a volume of 5600 cm³?

- (a) 11 cm (b) 12 cm (c) 9 cm (d) 13 cm

Ans: (a) 11 cm

$$\begin{aligned} \text{Volume of spherical vessel} &= 5600 \text{ cm}^3 = (4/3) * \pi * r^3 \\ \Rightarrow 5600 * 3 * 7 / (4 * 22) &= r^3 \Rightarrow 117600 / 88 = r^3 \Rightarrow r = 11 \text{ cm} \end{aligned}$$

56. Which of the following is the correct option expressing the volume of a hemisphere?

- (a) $(2/9) * \pi * r^3$ (b) $(2/3) * \pi * r^3$ (c) $(4/3) * \pi * r^3$ (d) $(4/9) * \pi * r^3$

Ans: (b) $(2/3) * \pi * r^3$

57. If two solid hemispheres of same base radii as $(3/2)r$, are joined together along their bases, then what would be the curved surface area of this new solid?

- (a) $9\pi r^2$ (b) $12\pi r^2$ (c) $16\pi r^2$ (d) $6\pi r^2$

Ans: (a) $9\pi r^2$

$$\begin{aligned} \text{Curved surface area of a solid hemisphere} &= 2 * \pi * ((3/2)r)^2 \\ \text{Thus, curved surface area of new solid} &= 2 * (2 * \pi * (9/4)r^2) = 9\pi r^2 \end{aligned}$$

58. What is the volume (in cubic cm) of a hemispherical bowl of radius 12 cm?

- (a) 1331π (b) 1420π (c) 1152π (d) 1519π

Ans: (c) 1152π

$$\text{Volume of hemisphere} = (2/3) * \pi r^3 = (2/3) * \pi * (12)^3 = 1152\pi$$

59. If the radius of a hemisphere is reduced to half, then, how many times is the volume of the new hemisphere to original hemisphere?

- (a) 1 / 8 times (b) 1 / 2 times (c) 1 / 4 times (d) 1 / 16 times

Ans: (a) 1 / 8 times

60. If the radius of a sphere is made five times, what will be the ratio of the volume of new sphere to the ratio of volume of original sphere?

- (a) 5: 1 (b) 25: 1 (c) 1: 625 (d) 125: 1

Ans: (d) 125: 1

$$\begin{aligned}\text{Volume of original sphere} &= (4/3) * \pi r^3 \\ \text{Volume of new sphere} &= (4/3) * \pi * (5r)^3 = (4/3) * 125\pi r^3 \\ \text{Ratio} &= \text{Volume of new sphere} / \text{Volume of original sphere} \\ &= ((4/3) * 125\pi r^3) / ((4/3) * \pi r^3) = 125:1\end{aligned}$$

61. A hemispherical bowl has a radius of 7 cm. How much cubic cm of water will this bowl contain if its half-filled?

- (a) 765.46 (b) 723.22 (c) 718.66 (d) 732.99

Ans: (c) 718.66

$$\text{Volume of hemisphere} = (2/3) * \pi r^3 = (2/3) * (22/7) * (7)^3 = 718.66 \text{ cm}^3$$

62. Two spheres of radius 2 cm each are melted and casted into a single sphere. What would be the radius of this new sphere?

- (a) 4 cm (b) $16^{1/3}$ cm (c) $12^{1/3}$ cm (d) 2 cm

Ans: (b) $16^{1/3}$ cm

$$2 * ((4/3) * \pi r^3) = (4/3) * \pi R^3 \Rightarrow 2 * ((4/3) * \pi (2)^3) = (4/3) * \pi R^3 \Rightarrow 2 * 8 = R^3 \Rightarrow R = 16^{1/3} \text{ cm}$$

63. What will be the volume of a sphere having radius as 21 cm?

- (a) 38,808 cm³ (b) 14,400 cm³ (c) 28,562 cm³ (d) 49,263 cm³

Ans: (a) 38,808 cm³

$$\text{Volume of sphere} = (4/3) * \pi r^3 = (4/3) * (22/7) * (21)^3 = 38,808 \text{ cm}^3$$

64. What will be the edge length of a cube having its total surface area equal to the volume of the cube?

- (a) 7 units (b) 3 units (c) 6 units (d) 1 unit

Ans: (c) 6 units

$$\text{Total surface area} = \text{Volume (for a cube)} \Rightarrow 6z^2 = z^3 \Rightarrow 6 = z$$

65. Three identical cubes each having a side length as x cm are joined end by end to form a cuboid. What will be the total surface area (in sq. cm) of this cuboid?

- (a) $14x^2$ (b) $16x^2$ (c) $18x^2$ (d) $12x^2$

Ans: (a) $14x^2$ cm²

$$\begin{aligned}\text{Total surface area of cuboid} &= 2lb + 2bh + 2hl \\ &= 2 * (3x) * (x) + 2 * (x) * (x) + 2 * (x) * (3x) = 6x^2 + 2x^2 + 6x^2 = 14x^2 \text{ cm}^2\end{aligned}$$

66. A cubical tank has an edge equal to 9 m. The amount of water in litres it will contain when it is three – fourth filled?

- (a) 439765 (b) 546750 (c) 678924 (d) 792679

Ans: (b) = 546750

$$a^3 = 9^3 = 729 \text{ m}^3 \Rightarrow \text{Three – fourth volume} = (3/4) * 729 = 546.75 \text{ m}^3 = 546750 \text{ L.}$$

67. What will be the volume of a rectangular tank having base area 84 m^2 and height as 5 m ?

- (a) 420 m^3 (b) 460 m^3 (c) 520 m^3 (d) 550 m^3

Ans: (a) 420 m^3

$$\text{Volume} = \text{Base area} * \text{Height} = 84 \text{ m}^2 * 5 \text{ m} = 420 \text{ m}^3$$

68. The lateral surface area of a cube is 81 m^2 . What will be the volume of this cube?

- (a) 625 m^3 (b) 590 m^3 (c) 729 m^3 (d) 780 m^3

Ans: (c) 729 m^3

$$\begin{aligned} \text{Lateral surface area} &= 81 \text{ m}^2 \Rightarrow a^2 = 81 \text{ m}^2 \Rightarrow a = 9 \text{ m} \\ \text{Volume} &= a^3 = (9\text{m})^3 = 729 \text{ m}^3 \end{aligned}$$

69. What will be the total surface area (in sq. m) of a cubical tank having side length as 6m ?

- (a) 195 (b) 256 (c) 274 (d) 216

Ans: (d) 216

$$\text{Total surface area of tank} = 6a^2 = 6 * (6)^2 = 216 \text{ m}^2$$

70. A cube of edge length 7 m is converted to a cuboid of height 7 m . What will be the area (in sq. m) of the base of this cuboid so formed?

- (a) 49 (b) 42 (c) 56 (d) 63

Ans: (a) 49

$$\begin{aligned} \text{Volume of cube} &= \text{Volume of cuboid} \Rightarrow 7 * 7 * 7 \text{ m}^3 = \text{Area of base} * \text{height} \\ &\Rightarrow 343 \text{ m}^3 = \text{Area of base} * 7 \text{ m} \Rightarrow 49 \text{ m}^2 = \text{Area of base} \end{aligned}$$

71. What would be the edge of a cube having volume equal to the volume of a cuboid having dimensions as $16 \text{ cm} * 4 \text{ cm} * 8 \text{ cm}$?

- (a) 0.6m (b) 0.02m (c) 0.08m (d) 0.4m

Ans: (c) 0.08 m

$$\begin{aligned} \text{Volume of cuboid} &= 16 \text{ cm} * 4 \text{ cm} * 8 \text{ cm} = 512 \text{ cm}^3 \\ \text{Volume of cube} &= s^3 \Rightarrow s^3 = 512 \text{ cm}^3 \Rightarrow s = 8 \text{ cm} \end{aligned}$$

72. A small cubical box measures 8 cm . What will be the volume (in cubic cm) of a packet containing 20 such boxes?

- (a) 6940 (b) 7680 (c) 7550 (d) 6540

Ans: (b) 7680 cm^3

$$\begin{aligned} \text{Volume of 1 small cuboidal box} &= 8 * 8 * 8 = 512 \text{ cm}^3 \\ \text{Volume of packet} &= 15 * \text{volume of small cuboidal box} = 15 * 512 \text{ cm}^3 = 7680 \text{ cm}^3 \end{aligned}$$

73. A larger cuboid having dimensions as $2\text{ m} \times 4\text{ m} \times 8\text{ m}$ is cut into a number of smaller cubes each having side length of 2 m . What will be the number of such small cubes?

- (a) 8 (b) 4 (c) 12 (d) 16

Ans: (a) 8

$$\text{Volume of larger cuboid} = 2\text{ m} \times 4\text{ m} \times 8\text{ m} = 64\text{ m}^3$$

$$\text{Volume of smaller cube} = 2 \times 2 \times 2\text{ m}^3 = 8\text{ m}^3$$

$$\text{Number of smaller cubes} = \text{Volume of larger cuboid} / \text{Volume of smaller cube} = 64\text{ m}^3 / 8\text{ m}^3 = 8$$

74. The volume of a solid sphere having a radius of $2r$ units length is (in cu. units).

- (a) $(32/3)\pi r^3$ (b) $(16/3)\pi r^3$ (c) $(8/3)\pi r^3$ (d) $(64/3)\pi r^3$

Ans: (a) $(32/3)\pi r^3$

75. If the ratio of the volumes of two solid sphere is $1:8$, the ratio of their curved surface area is

- (a) $1:2$ (b) $1:4$ (c) $1:8$ (d) $1:16$

Ans: (b) $1:4$

76. The whole surface area of a solid hemisphere with length of 7 cm radius is

- (a) $588\pi\text{ sq. m}$ (b) $392\pi\text{ sq. m}$ (c) $147\pi\text{ sq. m}$ (d) $98\pi\text{ sq. m}$

Ans: (b) $147\pi\text{ sq. m}$

$$\text{Whole surface area} = 3\pi (7)^2\text{ sq. cm} = 147\text{ sq. cm}$$

77. If the ratio of curved surface areas of two solid sphere is $16:9$, the ratio of their volumes is

- (a) $64:27$ (b) $4:3$ (c) $27:64$ (d) $3:4$

Ans: (a) $64:27$

78. If numerical value of curved surface area of a solid sphere is three times of its volume the length of radius is

- (a) 1 unit (b) 2 unit (c) 3 unit (d) 4 unit

Ans: (a) 1 unit

79. Volume of a sphere of radius $(r/2)$ units will be (in cu. units).

- (a) $(1/6)\pi r^3$ (b) $(4/3)\pi r^3$ (c) $(2/3)\pi r^3$ (d) $(1/3)\pi r^3$

Ans: (a) $(1/6)\pi r^3$

80. If numerical value of curved surface area of a solid sphere is equal to its volume, then the length of its radius is

- (a) 6 unit (b) 5 unit (c) 4 unit (d) 3 unit

Ans: (d) 3 unit

$$4\pi r^2 = \frac{4}{3}\pi r^3 \text{ or, } r = 3$$

81. The ratio of whole surface area of a solid sphere and solid hemisphere of same radius will be

- (a) 4: 3 (b) 3: 4 (c) 1: 2 (d) None of these

Ans: (a) 4: 3

$$\text{Ratio} = 4\pi (r)^2 : 3\pi r^2 = 4:3$$

82. The ratio of numerical values of volume and whole surface area of a sphere is 2:

1. Numerical value of radius is

- (a) 5 (b) 6 (c) 3 (d) 1

Ans: (b) 6

$$\frac{\frac{4}{3}\pi r^3}{4\pi r^2} = \frac{2}{1} \Rightarrow r = 6$$

83. Volume of a sphere is $(4/3)\pi r^3$ cu. units. The sphere is inscribed (in) a cube.

The ratio of volumes of the cube and the sphere is

- (a) 3: π (b) 4: π (c) 6: π (d) 8: π

Ans: (c) 6: π

84. The volume of a cylinder is given by:

- (a) $\pi r^2 h$ (b) $2\pi r h$ (c) $\pi r h^2$ (d) πr^2

Ans: (a) $\pi r^2 h$

85. If the length of radii of two solid right circular cylinders are in the ratio 2 : 3 and their heights are in the ratio 5: 3, the ratio of their lateral surface is

- (a) 2: 5 (b) 8: 7 (c) 10: 9 (d) 16: 9

Ans: (c) 10: 9

$$\text{Required ratio} = \frac{2\pi r_1 h_1}{2\pi r_2 h_2} = \frac{2}{3} \times \frac{5}{3} = 10 : 9$$

86. If volumes of two solid right circular cylinder are same and their height are in the ratio 1: 2, then the ratio of length of radii is

- (a) 1: $\sqrt{2}$ (b) $\sqrt{2}$: 1 (c) 1: 2 (d) 2: 1

Ans: (b) $\sqrt{2}$: 1

$$\pi r_1^2 h_1 = \pi r_2^2 h_2 \Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \frac{h_2}{h_1} = \frac{2}{1} \Rightarrow r_1 : r_2 = \sqrt{2} : 1$$

87. In a right circular cylinder, if the length of radius is halved and height is doubled, volume of cylinder will be

- (a) Equal (b) Double (c) Half (d) 4 times

Ans: (c) Half

88. If the length of radius of a right circular cylinder is doubled and height is halved, the lateral surface area will be

- (a) Equal (b) Double (c) Half (d) 4 times

Ans: (b) Double

89. Diameter and height of two solid right circular cylinders are 8 units and 3 (h – 1) units respectively and the volumes are equal, then h =?

- (a) 9 (b) 2 (c) 17 (d) 19

Ans: (d) 19

90. Volume of a cylindrical tank is 6160 cubic metres and radius is 14 metres. Depth is

- (a) 8 metres (b) 10 metres (c) 14 metres (d) 16 metres

Ans: (b) 10 metres

91. No. of coins with diameter 3 cm and 0.25 cm thickness from a cuboid of dimension 11 cm x 9 cm x 6 cm will be

- (a) 320 (b) 330 (c) 336 (d) 340

Ans: (c) 336

92. Area of the base and lateral surface area of a cylinder of radius 8 cm are equal. Height will be

- (a) 3 cm (b) 4 cm (c) 5 cm (d) 6 cm

Ans: (b) 4 cm

93. Volume of the hollow cylinder of outer radius R, inner radius r and height h will be

- (a) $\pi (R^2 + r^2) h$ (b) $\pi (R^2 - r^2) h$
(c) $(\pi/2) (R^2 - r^2) h$ (d) $(\pi/2) (R^2 + r^2) h$

Ans: (b) $\pi (R^2 - r^2) h$

94. If the length of radii of two solid right circular cylinders are in the ratio 2: 3 and their height in the ratio 5: 3, then the ratio of their volume is

- (a) 27: 20 (b) 20: 27 (c) 4: 9 (d) 9: 4

Ans: (b) 20: 27

$$\text{Required ratio} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \left(\frac{2}{3}\right)^2 \times \frac{5}{3} = \frac{4}{9} \times \frac{5}{3} = 20 : 27$$

95. If the dimensions of a cuboid in cm are 16 x 14 x 20, then its total surface area is.....

- (a) 4480 cm² (b) 1648 cm² (c) 824 cm² (d) 1740 cm²

Ans: (a) 4480 cm²

$$\text{TSA} = 2(lb + bh + hl) = 4480 \text{ cm}^2$$

96. The total surface area of hemisphere is 300 π , then find its radius.

- (a) 8 cm (b) 12 cm (c) 10 cm (d) 9 cm

Ans: (c) 10 cm

$$\text{TSA} = 3\pi r^2 = 300\pi \Rightarrow r = 10 \text{ cm}$$

97. Find the ratio of the volumes of a cylinder and cone having equal radius and equal height.

- (a) 1: 3 (b) 3: 1 (c) 2: 1 (d) 1: 2

Ans: (b) 3: 1

$$\frac{V_{cylinder}}{V_{cone}} = \frac{\pi r^2 h}{\frac{1}{3} \pi r^2 h} = 3: 1$$

98. The height of cone is 12 cm and radius is 5 cm then its slant height

- (a) 17 cm (b) 60 cm (c) 7 cm (d) 13 cm

Ans: (d) 13 cm

$$\text{For the cone, slant height } l = \sqrt{(r^2 + h^2)} = \sqrt{(5^2 + 12^2)} = 13 \text{ cm}$$

99. If the radius of a hemisphere is 7 cm, find its total surface area.

- (a) 452 cm² (b) 462 cm² (c) 472 cm² (d) 442 cm²

Ans: (b) 462 cm²

$$TSA = 3\pi r^2 = 3 \times \frac{22}{7} \times 7^2 = 462 \text{ cm}^2$$

100. A hemispherical bowl has a curved surface area of 150π cm². Find its radius.

- (a) 4√3 (b) 5√3 (c) 4√2 (d) 5√2

Ans: (b) 5√3

$$CSA = 2\pi r^2 = 150\pi \implies r^2 = 75 \implies r = \sqrt{75} = 5\sqrt{3} \text{ cm.}$$